

Preserving Future Choices in Agentic Semantic Communication

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Abstract—Agentic semantic communication increasingly lets an intelligent endpoint decide what to query, transmit, retain, or actuate as task state and communication opportunities evolve. Existing work has made these decisions task-aware through semantic value, goal-oriented state, context relevance, and long-horizon return. We identify a complementary failure mode motivated by pre-disaster emergency monitoring: a communication or information action can be highly useful now yet consume the only energy, contact, or observation-dependent resource needed to satisfy a later hard task requirement. We call the missing object *future-choice feasibility*: whether taking an action preserves at least one non-anticipative policy that can still complete all outstanding hard task semantics under future observations and shared communication opportunities.

We develop FUTURECHOICE, an exact-correct feasibility layer that combines replayable lower certificates, sound optimistic impossibility tests, set-level opportunity-conflict structure, persistent validity domains, and exact fallback. We deliberately separate method hardness from benchmark validity. A source-grounded executable emergency-communication benchmark retains ordinary-solved and negative cases rather than tailoring tasks to the method; an independent public deadline-and-battery mission then tests whether the same abstraction matters outside our simulator. On a frozen 100-instance hard-feasible N=10 cohort, FUTURECHOICE yields zero-tardiness completion for all four released heuristic policies, while depth-4 receding planning leaves residual failures; the first 21 instances are independently audited at 924 reached action frontiers with zero mismatch against exact continuation. Transferring the set-level conflict abstraction from the ASC formulation to a deadline-set MST bound reduces exact-correct search work by roughly 60% relative to our earlier bound on the frozen exact-audited cohort. These results position future-choice feasibility as a safety/completion complement to semantic value, not as another semantic value metric.

I. INTRODUCTION

Remote pre-disaster monitoring exposes a form of communication difficulty that is easy to miss when an action is judged only by its immediate semantic value. Challenged-networking systems already assume that connectivity can disappear for long intervals and that nodes must buffer, carry, and later forward information [1], [2], [3], [4]; geohazard deployments add strict monitoring continuity and warning requirements under constrained field infrastructure [5], [6], [7]. A landslide-monitoring node may have little battery, an uplink may disappear for hours, a gateway may temporarily lose its primary backhaul, and a warning phase may require denser sensing before an operator can restore connectivity. In this setting, a query, fallback transmission, or configuration change does more than reveal or deliver useful information. It also commits future energy, airtime, control opportunities, and queue capacity. An action that is locally legal—and even

immediately task-beneficial—can therefore destroy the only remaining way to satisfy a later monitoring obligation.

This problem sits naturally inside Agentic Semantic Communication (ASC). Recent goal-oriented and agentic communication systems decide what and when to acquire or transmit according to task relevance, Grade of Effectiveness, semantic information, context, or long-horizon return [8], [9], [10], [11]. Wireless-agent benchmarks likewise increasingly evaluate semantic/network reasoning and multi-step control [12], [13], [14]. These advances answer an increasingly rich version of “how useful is this communication action for the task?” They do not, however, explicitly represent whether the action leaves any *causal completion policy* that can still satisfy every outstanding hard task requirement after future observations arrive.

We study this missing feasibility primitive. For a legal history h_t and current action a , we define the *future-choice set* $\mathcal{F}(h_t, a)$ as the set of non-anticipative continuation policies that can still complete all active hard task obligations for every future observation branch compatible with the policy. The distinction is small but consequential. Consider a hidden state $H \in \{0, 1\}$ where querying H now consumes one unit of budget and reveals whether a later task requires B or C . One remaining query is sufficient after the observation: query B if $H = 0$, otherwise query C . A static worst-case reservation that unions both possible future obligations reserves $\{B, C\}$ simultaneously and rejects this valid contingent policy. The error is not an inaccurate semantic value. It is a representation error: mutually exclusive future task semantics were collapsed before the observation that separates them.

We therefore treat FUTURECHOICE as a feasibility layer around an existing ASC value policy rather than as another semantic score. It certifies feasible actions when a causal completion witness exists, prunes actions only through sound optimistic impossibility tests, and sends the unresolved remainder to exact fallback. The representation is observation-conditioned and set-level: mutually exclusive future obligations remain separated, while obligations that truly compete for the same future opportunity are reasoned about jointly.

A central methodological choice is to keep the motivating emergency problem unchanged. Our executable benchmark is frozen from field/standard requirements and retains ordinary-solved and negative cases rather than adding scarcity after seeing where FUTURECHOICE helps. It therefore acts as a *reality authority* and an anti-tailoring control, not as a generator of method-friendly hard cases.

To test whether the abstraction is benchmark-specific, we also use an independent public deadline-and-battery UAV mis-

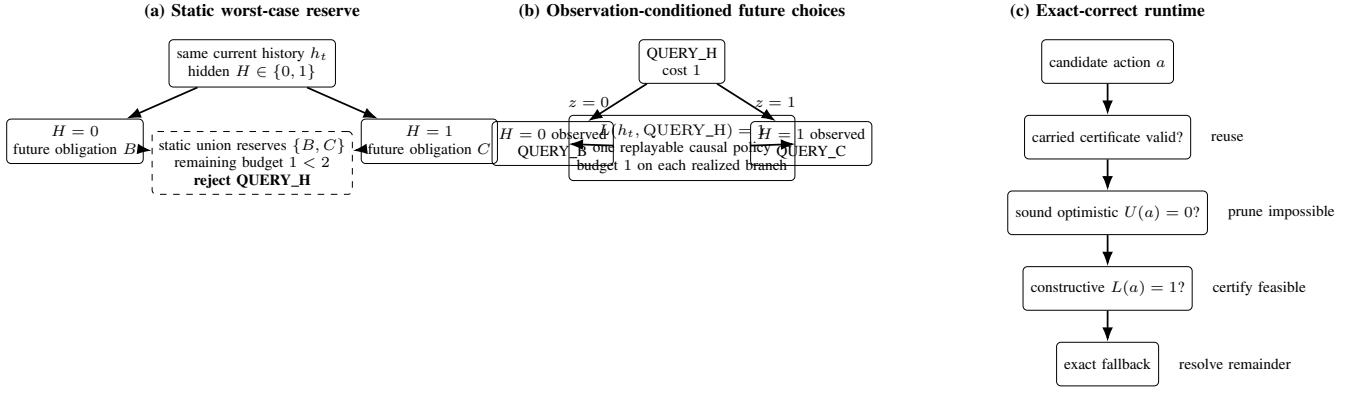


Fig. 1. Future-choice feasibility complements semantic value. (a) A static worst-case union can reject a valid contingent communication/query policy. (b) Conditioning future obligations on future observations preserves the causal branch structure. (c) FUTURECHOICE keeps exact correctness through carried certificates, sound optimistic pruning, constructive lower certificates, and exact fallback.

sion [15]. Its native action mask admits locally legal actions that can destroy an otherwise feasible zero-tardiness continuation. A method-independent 100-instance cohort exposes residual depth-4 failures that FUTURECHOICE removes; an embedded 21-instance exact-audited subset and a direct transfer of the ASC set-conflict abstraction provide separate correctness and attribution checks.

We make four contributions:

- **A missing ASC primitive.** We formulate *future-choice feasibility*, separating the semantic value of a current action from whether it preserves any non-anticipative way to complete remaining hard task semantics under future observations and shared opportunities.
- **An exact-correct incremental method.** We develop replayable lower certificates, sound optimistic upper tests, set-level conflict structure, certificate validity domains, and dependency-local invalidation under a common L/U +exact-fallback protocol.
- **Reality-grounded evaluation.** We construct a source-grounded executable benchmark for pre-disaster emergency communication and explicitly retain ordinary-solved and negative cases, separating benchmark validity from method hardness.
- **Cross-domain evidence.** We validate the same planning object on an independent deadline-and-battery mission, including an exact-audited correctness cohort, a 100-instance paired scale study, and a direct $B \rightarrow C$ attribution in which the ASC set-level conflict abstraction tightens the external domain's sound bound.

II. ASC PROBLEM SETTING AND EMERGENCY-SYSTEM MODEL

A. From emergency requirements to hard task semantics

Our primary scenario is pre-disaster mountain monitoring under intermittent communication and limited energy. We retain the system model already used by the source-grounded bench-

mark rather than introducing a separate planning abstraction. An operational task is

$$\tau = (\mathcal{N}_\tau, \mathcal{M}_\tau, \mathcal{P}_\tau, \mathcal{W}_\tau, \mathcal{D}_\tau, \mathcal{A}_\tau, \mathcal{S}_\tau), \quad (1)$$

which fixes target nodes and measurands, monitoring profiles, legal windows and deadlines, authorized effects/actions, and a source/scoring profile. A hard task-semantic obligation is

$$o = (i, m, r_o, [s_o, e_o], d_o), \quad (2)$$

requiring a valid sample of measurand m from node i in $[s_o, e_o]$ and delivery to the downstream center by d_o . This obligation is our executable form of goal-conditioned semantic relevance: it states what information must become available to the downstream task, by when, without prescribing a unique packet trajectory.

The physical communication state includes node batteries and caches, installed sensing/reporting configuration, access/backhaul state, gateway queues, and runtime state. We write

$$x_{t+1} = F(x_t, a_t, \xi_t; \phi), \quad (3)$$

where ξ_t contains exogenous harvest, temperature, wireless variation, outages, and task revisions; ϕ contains frozen deployment/protocol parameters. The agent does not observe x_t directly. At owner/location ℓ , legal observations are

$$z_t = H(x_{\leq t}, a_{< t}, \xi_{\leq t}, \ell), \quad (4)$$

and are materialized into an Evidence World $\mathcal{E}_t$ that preserves source, owner, generation/observation time, freshness, revision, and status. We use h_t to denote the legal history available to the decision process; hidden simulator truth is excluded.

This maps directly onto goal-oriented ASC: sensing nodes are semantic sources, evidence/query capabilities are semantic acquisition actions, communication/device capabilities are transmission or actuation actions, and the downstream obligation evaluator is the task-effectiveness authority. The key difference from a purely scalar task utility is that some semantics are *hard*: missing their release/window/deadline makes the episode operationally incomplete even if other messages have high semantic value.

B. Why value and feasibility are different

Let $Q_{\text{sem}}(h_t, a)$ be any existing semantic/task-value score. Our objective is not to replace it. We ask whether a is compatible with hard task completion. Let $\mathcal{W}(h_t)$ denote worlds consistent with the legal history, and let $\mathcal{O}(w, h)$ be the outstanding hard obligations in world w after history h . A continuation policy π is *non-anticipative*: two histories exposing the same legal observations must induce the same action until an observation distinguishes them.

We define

$$\mathcal{F}(h_t, a) = \{\pi : \forall w \in \mathcal{W}(h_t), \forall z \text{ reachable,} \\ \pi \text{ legally completes } \mathcal{O}(w, h)\}, \quad (5)$$

and the exact feasibility indicator

$$V^*(h_t, a) = \mathbf{1}[\mathcal{F}(h_t, a) \neq \emptyset]. \quad (6)$$

A value-driven ASC policy can therefore be wrapped as

$$\max_a Q_{\text{sem}}(h_t, a) \quad \text{s.t.} \quad V^*(h_t, a) = 1. \quad (7)$$

This formulation makes FUTURECHOICE a feasibility/safety complement to semantic value rather than another value metric.

III. FUTURE-CHOICE FEASIBILITY

A. Exact-correct L/U decomposition

Computing V^* exactly at every decision can be expensive. We maintain Boolean bounds

$$L_t(a) \leq V^*(h_t, a) \leq U_t(a), \quad L_t, U_t \in \{0, 1\}. \quad (8)$$

A replayable causal completion witness sets $L_t(a) = 1$. An infeasible *optimistic* relaxation sets $U_t(a) = 0$. Actions with $(L, U) = (0, 1)$ remain unresolved and fall back to exact search. The runtime order is deliberately conservative: carried certificate reuse, sound $U = 0$ pruning, constructive $L = 1$ certification, then exact fallback (Fig. 1).

a) Lemma 1 (lower-bound soundness).: If a certificate provides a legal, non-anticipative continuation that completes every active hard obligation on every compatible observation branch in its validity domain, then $L_t(a) = 1$ implies $V^*(h_t, a) = 1$. *Proof sketch.* The replayable certificate is itself an element of $\mathcal{F}(h_t, a)$.

b) Lemma 2 (upper-bound soundness).: Let $\tilde{P}(h_t, a)$ be a relaxation that only removes constraints or adds resources. If $\tilde{P}(h_t, a)$ is infeasible, then $U_t(a) = 0$ implies $V^*(h_t, a) = 0$. *Proof sketch.* Any real completion would also be feasible in the optimistic relaxation, giving a contradiction.

B. Set-level opportunity conflicts

Within a compatible branch, we construct a bipartite graph $G = (\mathcal{O}, \mathcal{R}, E)$ between hard obligations and future service opportunities. An edge indicates that a service opportunity can legally complete an obligation. Capacity-aware max-flow gives an optimistic matching; residual min-cuts expose Hall-deficient obligation subsets. The key object is therefore not whether each obligation has *some* candidate opportunity, but whether a

set of obligations jointly competes for the same scarce future opportunities.

This representation is observation-conditioned. Future obligations that belong to mutually exclusive observation branches are not statically unioned before the observation arrives. This distinction is what prevents the false-negative reservation pattern in Fig. 1.

C. Persistent validity and dependency-local invalidation

Each certificate stores a validity domain

$$D(c) = (\mathcal{W}_c, [b_{\min}, b_{\max}], \mathcal{D}_c, t_c^{\text{valid}}, \text{witness}), \quad (9)$$

where $\mathcal{D}_c$ contains the obligations, communication opportunities, and pending execution/evidence events on which the certificate depends. Observation narrowing can shrink $\mathcal{W}_c$ without invalidating the witness. Conversely, a resource or time event invalidates only certificates whose dependency domain changes.

a) Lemma 3 (dependency-disjoint preservation).: If an event leaves a certificate's support/resource/time validity domain unchanged, modifies no dependency in $\mathcal{D}_c$, and introduces no new legal opportunity edge connecting the reused component to another component, the certificate remains valid. If the last condition cannot be guaranteed, the runtime conservatively repartitions or fully rebuilds rather than reusing the certificate.

This lemma captures a systems distinction that ordinary history caches miss: *history changed* does not necessarily imply *future-feasibility structure changed*.

D. A transferable optimistic bound

The external UAV adapter instantiates the same set-level idea using deadline sets. For each deadline threshold d , let S_d be unserved customers due by d . Any route prefix serving S_d must form a connected walk spanning the current position and S_d , whose length is at least their Euclidean minimum spanning tree. Thus

$$t + \frac{\text{MST}(\{x_t\} \cup S_d)}{v} > d \implies U_t(a) = 0. \quad (10)$$

Battery limits, charging detours, and depot return are omitted, so the test is optimistic.

a) Lemma 4 (deadline-set MST soundness).: If the deadline-set MST inequality above fails for any d , no real route can serve all customers in S_d by d . Hence pruning the action is sound.

IV. SOURCE-GROUNDED EMERGENCY EVALUATION

Our benchmark is not constructed around FUTURECHOICE. It is derived independently from field, standard, and procurement requirements for pre-disaster emergency monitoring. This design choice is important: many realistic tasks are in fact saturated by ordinary mechanisms, and we retain them as conformance or interactive-decision cases rather than removing them for lack of method headroom.

The benchmark uses three strata: *Operational-Conformance*, *Interactive-Decision*, and *Future-Choice-Stress*. The first validates source/task/evaluator correctness; the second exercises

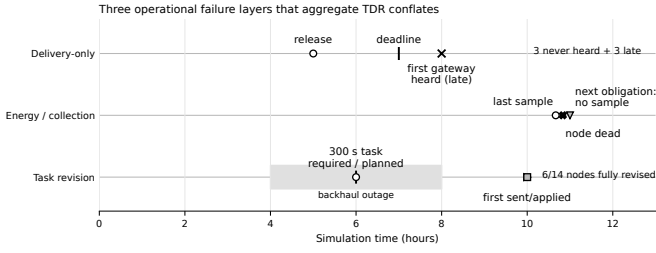


Fig. 2. Three rule-selected failure cases from the frozen deterministic test. Delivery-only, energy/collection, and task-revision failures occupy different points in the sensing–gateway–center lifecycle even when an aggregate delivery metric hides the distinction.

sequential actions and feedback but may be ordinary-solved; only the third is permitted to support a method-necessity claim. The current paper-facing release contains seven executable T1 surfaces, a frozen 200/200/150 train/dev/test coordinate split, execution-evaluator mutation tests, and a frozen baseline protocol. The 150-coordinate deterministic test produces 750 online baseline rows and 210 bounded oracle diagnostics. A frozen 30-coordinate $\times$ two-context-mode model-spectrum subset is evaluated separately and is not used for the FUTURECHOICE claim.

Figure 2 illustrates why final-state/lifecycle evaluation matters. In the delivery-only case, sensing succeeds but some samples never reach the gateway and others arrive after the deadline. In the energy case, the node dies before the next required sample exists. In the task-revision case, the denser task becomes active during a backhaul outage and only a subset of nodes installs both required settings. These are different operational failures and should not be collapsed into one “bad communication” score.

V. EVALUATION METHODOLOGY

We organize evaluation around five research questions rather than around code modules.

RQ1: Is the emergency benchmark valid, and are realistic tasks all hard? We audit source-to-task provenance, execution-evaluator mutations, ordinary deterministic baselines, and lifecycle failures. Negative/saturated cases are retained.

RQ2: When do scalar value and static reservation fail? Controlled ASC families isolate observation-conditioned obligations and shared opportunity conflicts, comparing value/static-union views with exact contingent feasibility.

RQ3: Can the structure be maintained exactly and incrementally? We compare fresh exact, ordinary persistent exact, dependency-cache exact, and incremental conflict certificates; a separately frozen structural holdout changes graph motifs rather than merely increasing size.

RQ4: Does preserving future choices change outcomes outside our benchmark? We use the public deadline/battery UAV environment. The main exact-correct cohort has 21 method-independent constructive hard-feasible seeds and 924 independently checked frontiers. A confirmatory scale panel extends to 100 constructive hard-feasible seeds, while a separate

50-episode panel exactly matches the external repository’s published evaluation seed protocol.

RQ5: Do the ASC and UAV results instantiate the same abstraction? We attribute the external-domain search reduction to the set-level conflict bound and route both domains through the same generic FUTURECHOICE orchestration protocol.

All exact or large-scale experiments follow a resource-safe execution contract; timeout or memory limits are reported as unresolved computation, never as infeasibility. We report task quality and search/recomputation work separately and do not translate heterogeneous internal work units into wall-clock speedup claims.

A. Cohorts, baselines, and correctness authorities

The experiments deliberately separate *correctness* from *scale*. RQ2 uses bounded synthetic ASC families because every action frontier can be checked against exact contingent feasibility. RQ3 uses both the development ladder ($C \in \{2, 4, 8, 16\}$ disconnected components) and a separately frozen four-motif structural holdout; incremental certificates are compared against fresh structural rebuilds at every event. Exact strong controls are ordinary persistent AND–OR search and a dependency-projected persistent exact cache. Resource-limited exact cells remain unresolved rather than being dropped or reclassified.

RQ4 uses three external cohorts with distinct roles. The 21-instance $N=10$ cohort is the *correctness core*: all 924 reached action frontiers are independently checked against exact continuation. The 100-instance cohort is a *scale-confirmatory panel*: every layout is selected before FUTURECHOICE evaluation by an ordinary released heuristic that already provides a zero-tardiness complete route, and its first 21 instances are exactly the correctness core. The third panel reproduces the external repository’s own 50-episode evaluation seed protocol and is used only for distribution/scale context; it is not used to manufacture hard-feasible cases. This nesting lets us increase statistical scale without repeating every expensive exact audit or weakening the correctness authority.

The baseline ladder is chosen to eliminate progressively stronger ordinary explanations. In ASC we compare scalar/value scheduling, static worst-case reservation, generic constrained exact search, ordinary persistent exact, dependency-cache exact, and the conflict frontier. In the external mission we compare the released native heuristics, depth- k receding feasibility masks (with depth 4 as the strongest fixed-horizon control used in the main result), pure exact continuation, and the exact-correct L/U +fallback layer. The set-MST experiment changes only the optimistic U bound, holding the heuristic policy, exact authority, and cohort fixed.

B. Metrics and statistical protocol

For binary task outcomes we report raw paired counts and Wilson 95% intervals. On the 100-instance paired cohort, FutureChoice–baseline differences in zero-tardiness, completion, infeasible-episode incidence, tardiness, and energy are summarized with a fixed-seed paired bootstrap; zero-tardiness

TABLE I
FROZEN EXPERIMENT DESIGN. SCALE EXPERIMENTS AND CORRECTNESS EXPERIMENTS ARE DELIBERATELY SEPARATED: LARGE PANELS EXTEND STATISTICAL STABILITY, WHILE EXACT-FRONTIER AUDITS REMAIN ON BOUNDED PREREGISTERED COHORTS.

RQ	Evidence family and frozen scale	Correctness / control gate
RQ1	Emergency benchmark validity. 150 frozen coordinates; 750 online rows; 210 oracle diagnostics	source/evaluator audit + frozen deterministic test
RQ2	Conditional ASC representation. 20 exact-audited cells; 18 cells; 15 nontrivial	exact contingent feasibility vs static union/reserve
RQ3	Incremental correctness / structure. C=2,4,8,16; 4 unseen B6 graph motifs	fresh rebuild + persistent/dependency exact controls
RQ4	External mission outcomes. 21 constructive seeds x 4 policies; 924 exact frontier checks; 100 of 126 constructive seeds from scan 0..999; first 21 exactly equal C2 exact-audited cohort	924 exact frontiers + paired depth-4 / scale statistics
RQ5	Cross-domain transfer. Same exact-audited N=10 cohort + generic engine parity	set-MST attribution + shared L/U orchestration

discordance is additionally tested with the exact two-sided McNemar test. Structural experiments report exact frontier equality, component recomputation/reuse, branch-frontier builds, and exact-search expansions. These are intentionally kept as separate work units. Timing measurements appear only as implementation diagnostics unless the compared methods execute the same work unit under the same runtime contract.

VI. RESULTS

A. RQ1: Reality first—many emergency tasks are ordinary-solved

The benchmark study finds that several plausible “agentic” mechanisms are absorbed by ordinary controls: persistent-energy commitment can be handled by horizon-aware resource admission; shared backup stress can be absorbed by packing/max-coverage rules; and source-valid warning handoff tasks do not justify inventing an unobserved cross-stage contact quota. These negative results support the benchmark’s role as a method falsifier rather than a method generator.

Table II shows why we avoid labeling the entire benchmark “Agentic-hard.” O1 is nearly saturated by an ordinary AoI policy, while O3 attains high mean TDR under several simple rules yet still reaches full routine success on only three of 30 coordinates. O2 and O6 remain far from saturation because task revision, collection, and delivery failures occur together. O4 exposes a different trade-off: EA-AoI achieves high TDR but still incurs occasional collection and energy-exhaustion failures. The benchmark therefore spans conformance, saturated interactive cases, and genuinely unresolved workflows instead of filtering for a single preferred algorithmic mechanism.

B. RQ2–RQ3: Conditional structure is necessary and maintainable

Static union reservation produces false negatives whenever future obligations belong to mutually exclusive observation branches. The shared-opportunity stress family preserves a local per-branch backup requirement while the static union grows with the number of branches. Persistent conditional domains avoid rebuilding inactive branch frontiers, and active-branch events invalidate only dependency-intersecting components.

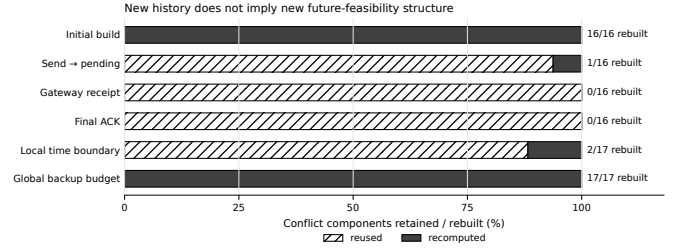


Fig. 3. Dependency-local invalidation for a 16-component revealed branch. A new gateway receipt or final ACK changes history but requires no conflict-component rebuild once the accepted send already secures the obligation; a global backup-budget change invalidates all dependent components.

We additionally froze a structural holdout *before* execution, replacing the development pattern of equal disconnected components with four unseen graph motifs: asymmetric components, a bridged pair, a chain that splits as shared opportunities disappear, and mixed local/global invalidation. Across all four motifs and every frozen event, the incremental structural snapshot is canonically identical to a fresh full rebuild. The holdout also exposes the intended safety boundary: bridge removal expands or splits the affected dependency scope, while a global satellite-budget change invalidates all backup-dependent components. Under a separate 512 MiB/30 s exact-control budget, six events are resolved by fresh, ordinary persistent, and dependency-cache exact; all six yield identical action frontiers. The remaining exact cells are reported only as unresolved computation.

The mixed local/global motif also uncovered an implementation bottleneck in our optimistic physical upper test: combined terrestrial/satellite assignment was implemented by exponential DFS although the constraint is an integral max-flow with a shared satellite-budget pool. Replacing the DFS with an exhaustively cross-checked max-flow leaves all prior non-timing results unchanged and makes the structural holdout tractable. We treat this as an implementation correction, not as a planner-level wall-clock superiority claim.

C. RQ4: External task-quality headroom beyond depth-4

On the frozen 100-instance hard-feasible N=10 cohort, FUTURECHOICE obtains zero-tardiness completion on all four

TABLE II

DETERMINISTIC LAYER-1 LANDSCAPE ON FROZEN TEST COORDINATES. THE TABLE REPORTS THE BEST ORDINARY BASELINE BY MEAN TDR ONLY; TIES ARE RETAINED RATHER THAN BROKEN POST HOC. HIGH TDR DOES NOT IMPLY FULL OPERATIONAL SUCCESS, AND FAILURE CLASSES REMAIN SEPARATELY VISIBLE.

Task	Best ordinary baseline	Mean TDR	Full success	Residual failures
O1	AoI	0.999	27/30	delivery deadline
O2	Det.-comply/Mission-comply/Mission-sustain	0.326	0/30	collection, delivery deadline, task revision
O3	Det.-comply/EDF/MaxCov/Local	0.979	3/30	delivery deadline
O4	EA-AoI	0.994	17/30	collection, delivery deadline, energy exhaustion
O6	EA-AoI	0.279	0/30	collection, delivery deadline, task revision

TABLE III

ASC REPRESENTATION AND MAINTENANCE EVIDENCE. WORK UNITS ARE INTENTIONALLY REPORTED WITHOUT CONVERTING COMPONENT RECOMPUTATIONS INTO EXACT-SEARCH EXPANSIONS.

Mechanism	Result	Unit / boundary
Static union	15/15 false negatives	representation failure
Observation narrowing	192 to 32 builds	persistent domains
Fresh ordered exact	866	exact expansions
Persistent exact	246	exact expansions
Dependency-cache exact	246	separator replay=0
Conflict frontier	36 recompute / 62 reuse	different work unit

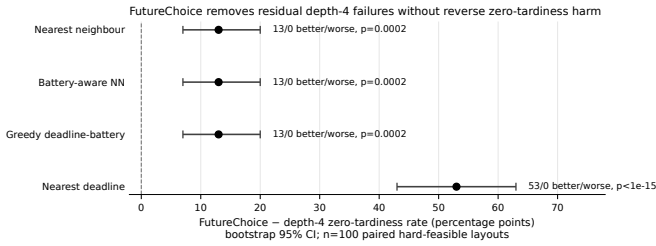


Fig. 4. Paired zero-tardiness gain of FUTURECHOICE over depth-4 receding planning on 100 preselected hard-feasible N=10 layouts. Error bars are paired bootstrap 95% confidence intervals; annotations show favorable/adverse discordance and exact McNemar tests.

released heuristic policies. Depth-4 receding planning still leaves residual failures, especially for the nearest-deadline policy. Importantly, the first 21 instances are exactly the previously frozen exact-audited cohort; thus correctness and scale are nested rather than being established on unrelated samples.

D. RQ5: The set-level abstraction transfers

Replacing an individual-deadline/global-MST optimistic bound with the ASC-inspired deadline-set conflict bound leaves task outcomes unchanged while reducing exact fallback and total exact-correct search work on the frozen exact-audited cohort.

The resulting correctness-compute picture is deliberately not a universal dominance claim. Within exact-correct methods, the set-level bound dominates the older basic bound and pure exact-mask point under the declared search-state proxy. Once depth-4 is included, depth-4 and FUTURECHOICE are both non-dominated: depth-4 is cheaper but leaves residual hard-

TABLE IV

N=10 SCALE-CONFIRMATORY OUTCOMES ON 100 METHOD-INDEPENDENT HARD-FEASIBLE LAYOUTS. THE FIRST 21 ARE THE EXACT-AUDITED COHORT.

Policy	Native zero	Depth-4 zero	Future zero	Depth-4 comp.	Future comp.	Depth-4 infeas.	Future infeas.
Nearest neighbour	49/100	87/100	100/100	100/100	100/100	0	0
Nearest deadline	12/100	47/100	100/100	69/100	100/100	41	0
Greedy deadline-battery	79/100	87/100	100/100	98/100	100/100	3	0
Battery-aware NN	49/100	87/100	100/100	100/100	100/100	0	0

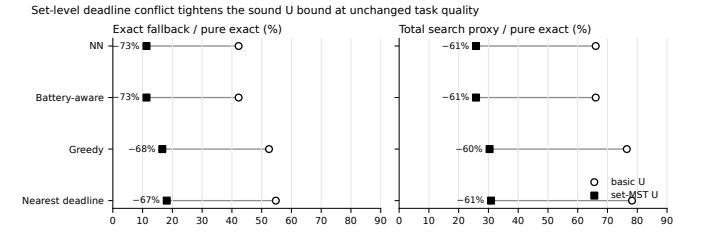


Fig. 5. B→C attribution. The deadline-set MST bound reduces exact fallback and total exact-correct search work relative to the earlier basic optimistic bound at unchanged task quality.

feasibility failures, whereas FUTURECHOICE buys exact-correct completion at higher search cost.

VII. RELATED WORK

a) *Challenged networking, freshness, and resource-aware delivery.*: DTN and scheduled-contact networking established store-carry-forward communication under intermittent connectivity and finite contacts [1], [2], [4], [16]. Queue management, custody/expiry semantics, and satellite-assisted IoT further make delivery state persistent across contacts [17], [5]. Separately, Age of Information and Value of Information quantify freshness and task relevance [18], [19], [20], while energy-harvesting and deadline-aware communication explicitly couple update timing to resource availability [21], [22], [23], [24]. We inherit these physical concerns; our additional object is whether a current communication action preserves a complete future policy for all outstanding hard task semantics rather than only improving freshness, expected value, or one-step deadline slack.

b) *Semantic and goal-oriented communication.*: Semantic communication has progressively moved from representation efficiency toward task effectiveness and goal-oriented decision making [25], [26]. Pull-based scheduling explicitly optimizes what and when to query under long-term semantic effectiveness and query cost [8]; logical goal-oriented communication builds

decision-relevant semantic states [10]; world-model semantic communication estimates long-horizon counterfactual contribution to physical-AI return [9]; and wireless context engineering treats context selection and injection as a wireless resource-allocation problem [11]. FUTURECHOICE complements these scalar or representational objectives with a hard completion-feasibility layer.

c) Agentic wireless systems and benchmarks.: Recent work brings foundation models and agents into wireless control, tool use, and network management [27], [28]. 6G-Bench and WirelessBench provide broad semantic/network reasoning benchmarks, while α^3 -Bench evaluates safety, robustness, and efficiency for UAV agents [12], [13], [14]. DORA provides expert-authored operational disaster tasks and replayable tool trajectories [29]. Our benchmark instead traces emergency-communication requirements into executable communication lifecycles and final hard-obligation outcomes; its role in this paper is to ground and falsify the method rather than to make every task algorithmically hard.

d) Active information acquisition and constrained planning.: Decision-region determination, action-sufficient representations, stochastic Boolean evaluation, and adaptive probing all study how much information is needed for a decision [30], [31], [32], [33]. Active measuring further models delayed negative effects of measurement itself [34]. Our distinction is that the controlled communication action changes a set of future hard obligations/opportunities under future observations; we maintain executable completion certificates and their validity domains rather than only a measurement value or stopping criterion.

VIII. DISCUSSION AND LIMITATIONS

Our current implementation establishes exact action-feasibility and structured work reduction, not universal wall-clock superiority. Ordinary persistent exact search remains a strong systems baseline, and depth-4 receding planning is a cheaper operating point on some external policies. The emergency benchmark’s Future-Choice-Stress stratum may remain empty: method evidence comes from the direct ASC formulation and independent external mission rather than from retrofitting scarcity into the benchmark. The N=15 routing result is a bounded probe, not a distribution-level result. The source/task/evaluator audit is internal rather than independent domain-expert validation. The frozen LLM benchmark subset uses one provider/model family and remains a model-spectrum diagnostic, not a contribution to the FUTURECHOICE method claim.

IX. CONCLUSION

Agentic semantic communication should not be defined by adding an LLM or another semantic-value score. The difficult regime arises when current communication and information actions change which future hard task completions remain possible. FUTURECHOICE makes this regime explicit through observation-conditioned completion sets, replayable feasibility certificates, sound optimistic impossibility tests, and

dependency-local maintenance. A source-grounded emergency benchmark keeps the method tied to legitimate operational semantics, while an independent deadline-and-battery mission shows that the same planning object matters outside the motivating simulator. Our results support future-choice feasibility as a complementary safety layer around existing ASC value policies.

APPENDIX A PROOF DETAILS

A. Replayable lower certificates

Consider a current legal history h_t and action a . A lower certificate stores a continuation witness π_c together with a declared validity domain $D(c)$. By construction, replay validation checks that (i) every action in π_c is legal under the observation history on which it is taken, (ii) the policy branches only after an observation distinguishes histories, and (iii) every hard obligation active on each covered compatible branch is completed before its operational deadline. Therefore $\pi_c \in \mathcal{F}(h_t, a)$ whenever $D(c)$ holds, proving $L_t(a) = 1 \Rightarrow V^*(h_t, a) = 1$.

The important restriction is non-anticipativity. A world-indexed open-loop trace is not a valid lower certificate unless the traces agree on every prefix with the same legal observations. This is why a perfect-information physical solution is an upper-bound aid, not a constructive lower certificate.

B. Optimistic upper tests

Let $P(h_t, a)$ denote the real continuation-feasibility problem and let $\tilde{P}(h_t, a)$ be a relaxation obtained only by deleting constraints, shortening delays, increasing resources, or granting additional information. Every real feasible continuation is then feasible in $\tilde{P}$. Hence

$$\tilde{P}(h_t, a) \text{ infeasible} \implies P(h_t, a) \text{ infeasible}, \quad (11)$$

which establishes the soundness of $U_t(a) = 0$. The converse does not hold: an optimistic relaxation may be feasible even when no causal real policy exists.

C. Dependency-local preservation

Let c be a component certificate with dependency set $\mathcal{D}_c$, compatible-world support $\mathcal{W}_c$, resource interval, and validity boundary t_c^{valid} . Suppose a new event changes the global history from h to h' . If (i) the event preserves $\mathcal{W}_c$ or only narrows it, (ii) no state read by the certificate witness changes, (iii) the current time remains inside the validity interval, and (iv) no new legal future-opportunity edge joins the reused component to a previously separate component, then every obligation–opportunity relation used by the certificate is unchanged. The previous witness or Hall-conflict structure therefore remains valid.

Condition (iv) is the partition-safety boundary. The current transition model is monotone in future opportunity edges: time and resource use remove edges, while execution does not create an unseen terrestrial opportunity between previously unrelated components. The implementation nevertheless checks for

overlap between rebuilt and reused slot sets; if the assumption is violated, it conservatively repartitions the whole world rather than reuse a potentially unsound certificate.

D. Deadline-set MST upper bound

For the external routing adapter, fix a deadline threshold d and let S_d contain all unserved customers whose deadlines are at most d . Any route that serves every node in S_d before d induces a connected walk spanning the current UAV position and all points in S_d . The length of any such connected walk is at least the Euclidean minimum spanning tree length over those points. Therefore

$$t + \frac{\text{MST}(\{x_t\} \cup S_d)}{v} > d \quad (12)$$

is sufficient to prove that no real route can meet all deadlines in S_d . The test ignores battery limits, charging detours, future return-to-depot requirements, and other constraints; all omissions make the relaxation more optimistic, so a negative result remains sound.

APPENDIX B

STRUCTURAL HOLDOUT AND RESOURCE-BOUNDED EXACT CONTROLS

The B6 holdout was frozen before execution and changes graph topology rather than only scaling the number of components. H1 uses asymmetric disconnected groups; H2 introduces a shared terrestrial bridge; H3 creates a chain whose connected component splits as bridge opportunities disappear; H4 interleaves local invalidation with a global shared satellite-budget dependency.

All four motifs pass the structural equality gate. The exact strong-control audit is intentionally resource-bounded: each motif-baseline sequence runs in an isolated process with a 512 MiB address-space ceiling and 30 s wall timeout. Fresh, ordinary persistent, and dependency-cache exact all resolve on six common events and agree on the full action frontier on all six. Other sequences terminate because of memory or time limits and remain `UNRESOLVED_COMPUTATION`; they do not contribute to feasibility or method-superiority claims.

The holdout also revealed a systems implementation issue. The original optimistic physical upper test solved combined terrestrial/satellite assignment by recursive enumeration. The constraint is instead an integral flow network: obligations connect to concrete unit-capacity slots, terrestrial slots connect directly to the sink, and satellite slots connect through a shared pool whose capacity equals the remaining backup budget. We retain the historical DFS as a test reference and exhaustively compare it to the production max-flow implementation on all candidate-subset combinations of a three-obligation/four-slot test space and four satellite budgets. All cases agree; existing F1–F3 tracked outputs are unchanged except for timing fields.

APPENDIX C

EXTERNAL COHORT CONSTRUCTION AND STATISTICS

The external N=10 study separates correctness from statistical scale. We first scan seeds using only the four released

ordinary heuristics. A seed enters the constructive pool if at least one heuristic already completes all customers, returns to the depot, incurs zero tardiness, and takes no infeasible action. FutureChoice and exact continuation are not used for selection. The first 21 constructive seeds form the exact-audited cohort; the first 100 form the scale-confirmatory cohort, so the correctness core is an exact prefix of the scale panel rather than a different sample.

For every reached action frontier in the 21-instance core, we compare the FutureChoice mask against an independent exact-continuation mask; the 924 reached frontiers have zero mismatch. The 100-instance panel is used for paired outcome statistics rather than repeating every exact audit. For binary outcomes we report Wilson intervals. Paired rate differences, tardiness changes, and energy changes use a fixed-seed paired bootstrap; zero-tardiness discordance additionally uses the exact two-sided McNemar test.

The external-paper-matched 50-episode panel serves a different purpose. Its layout seeds are generated by the released repository’s own evaluation protocol, namely the sequence produced by `np.random.default_rng(999)` for 50 episodes. Only six of those 50 layouts have an ordinary constructive zero-tardiness witness. We therefore run FutureChoice only on those six for hard-feasibility comparison and report the full 50-episode native/depth-4 results as distribution context. We do not interpret an unlabelled layout as hard-infeasible merely because an ordinary heuristic fails to produce a witness.

APPENDIX D

BENCHMARK VALIDITY AND MODEL-SPECTRUM BOUNDARY

The emergency benchmark is frozen independently of the method. Source/task/evaluator construction, the 200/200/150 coordinate split, deterministic baselines, execution-evaluator mutation tests, and the model-spectrum subset are all tracked as separate artifacts. Operational-Conformance and Interactive-Decision cases remain in the release even when ordinary rules solve them. Future-Choice-Stress is allowed to be empty; an empty stress stratum weakens a method-necessity claim inside the emergency benchmark but does not invalidate the benchmark itself.

The frozen LLM subset contains 30 test coordinates evaluated under two context modes. It is explicitly a model-spectrum diagnostic. Its role is to test whether the benchmark exposes model/runtime behavior that differs from the deterministic baseline landscape; it is not used as evidence that FutureChoice requires an LLM or that the proposed method improves language-model reasoning. Runtime failures, replay failures, and token/call/latency costs are preserved rather than silently dropped.

REFERENCES

- [1] K. Fall, “A delay-tolerant network architecture for challenged internets,” in *Proceedings of ACM SIGCOMM*, 2003, pp. 27–34.
- [2] S. Jain, K. Fall, and R. Patra, “Routing in a delay tolerant network,” in *Proceedings of ACM SIGCOMM*, 2004, pp. 145–158.

TABLE V

B6 STRUCTURAL HOLDOUT. “INC.=FRESH” MEANS EVERY EVENT-LEVEL INCREMENTAL SNAPSHOT IS CANONICALLY EQUAL TO A FULL FRESH REBUILD. EXACT COMPARISONS INCLUDE ONLY EVENTS WHERE ALL THREE RESOURCE-BOUNDED EXACT CONTROLS RESOLVE.

Motif	Events	Inc.=fresh	Recomp./reuse/fallback	Exact match/resolved	Unresolved seq.
H1 ASYMMETRIC_DISJOINT	6	all	15/15/0	0/0	3
H2 BRIDGED_PAIR	5	all	13/6/0	4/4	1
H3 CHAIN_OVERLAP_SPLIT	5	all	17/10/0	2/2	2
H4 MIXED_LOCAL_GLOBAL	6	all	36/15/0	0/0	3

- [3] K. Fall, S. Burleigh, A. Hooke *et al.*, “Delay-tolerant networking architecture,” IRTF / RFC Editor, Tech. Rep. RFC 4838, 2007.
- [4] IETF Delay-Tolerant Networking Working Group, “Bundle protocol version 7,” IETF / RFC Editor, Tech. Rep. RFC 9171, 2022.
- [5] Y. Ren, A. Gamage, L. Liu, M. Li, S. Chen, Y. Dong, and Z. Cao, “SateRIoT: High-performance ground-space networking for rural IoT,” in *Proc. 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom ’24)*, 2024.
- [6] Ministry of Natural Resources of the People’s Republic of China, “DZ/T 0450–2023, communication-layer specification for geological-disaster monitoring (clauses 7.4.2.3, 7.4.3.2, 7.4.3.5),” Industry standard of the People’s Republic of China, Tech. Rep., 2023.
- [7] —, “DZ/T 0460–2023, technical specification for geological-disaster monitoring (clauses 5.3.3, 5.3.6, 5.3.7, 8.4.1, 8.4.2),” Industry standard of the People’s Republic of China, Tech. Rep., 2023.
- [8] P. Agheli, N. Pappas, and M. Kountouris, “Pull-based query scheduling for goal-oriented semantic communication,” *arXiv preprint arXiv:2503.06725*, 2025.
- [9] L. Wang, T. Shui, W. Saad, and P. Adjakple, “World model-enabled causal digital twins for semantic communications in physical AI systems,” *arXiv preprint arXiv:2605.16547*, 2026.
- [10] A. F. Saz and F. Fekri, “Goal-oriented semantic communication for logical decision making,” *arXiv preprint arXiv:2604.19614*, 2026.
- [11] C. Zhao, J. Wang, Y. Xu, G. Sun, D. Niyato, Z. Li, A. Jamalipour, and D. I. Kim, “Wireless context engineering for efficient mobile agentic AI and edge general intelligence,” *arXiv preprint arXiv:2602.07321*, 2026.
- [12] M. A. Ferrag, A. Lakas, and M. Debbah, “6G-Bench: An open benchmark for semantic communication and network-level reasoning with foundation models in AI-native 6G networks,” *arXiv preprint arXiv:2602.08675*, 2026.
- [13] J. Tong, F. Liu, L. Xv, S. Lu, K. Li, Y. Zhang, Y. Song, Z. Xue, and J. Zhang, “WirelessBench: A tolerance-aware LLM agent benchmark for wireless network intelligence,” *arXiv preprint arXiv:2603.21251*, 2026.
- [14] M. A. Ferrag, A. Lakas, and M. Debbah, “ α^3 -Bench: A unified benchmark of safety, robustness, and efficiency for LLM-based UAV agents over 6G networks,” *arXiv preprint arXiv:2601.03281*, 2026.
- [15] M. Dehghani and A. Jain, “Attention-based deep reinforcement learning for UAV routing under deadline and battery constraints,” *IIEE Transactions*, 2026, under review; companion code released by the authors.
- [16] Consultative Committee for Space Data Systems, “Schedule-aware bundle routing,” CCSDS, Tech. Rep. CCSDS 734.3-B-1, Blue Book, Issue 1, 2019.
- [17] S. Iranmanesh, “A novel queue management policy for delay-tolerant networks,” *EURASIP Journal on Wireless Communications and Networking*, vol. 2016, no. 88, pp. 1–23, 2016.
- [18] S. Kaul, R. Yates, and M. Gruteser, “Real-time status: How often should one update?” in *2012 Proceedings IEEE INFOCOM*, 2012, pp. 2731–2735.
- [19] R. D. Yates, Y. Sun, D. R. Brown, S. K. Kaul, E. Modiano, and S. Ulukus, “Age of information: An introduction and survey,” *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 5, 2021.
- [20] S. Wang, Q. Liu, P. U. Abara, J. S. Baras, and S. Hirche, “Value of information in networked control systems subject to delay,” in *2021 60th IEEE Conference on Decision and Control*, 2021, pp. 1275–1280.
- [21] B. T. Bacinoglu, Y. Sun, E. Uysal, and V. Mutlu, “Optimal status updating with a finite-battery energy harvesting source,” *arXiv preprint arXiv:1905.06679*, 2019.
- [22] A. Zakeri, M. Moltafet, and M. Codreanu, “Semantic-aware sampling and transmission in energy harvesting systems: A POMDP approach,” *arXiv preprint arXiv:2311.06522*, 2023, preliminary results at Asilomar 2023 and IEEE GLOBECOM 2023.
- [23] M. Costa, M. Codreanu, and A. Ephremides, “On the age of the information in status update systems with packet management,” *IEEE Transactions on Information Theory*, vol. 62, no. 4, pp. 1897–1910, 2016, conference version in *Proc. IEEE ISIT 2014*, pp. 1583–1587.
- [24] C. ping Kam, S. Kaya, and A. Ephremides, “Controlling the age of information: Buffer size, deadline, and packet replacement,” in *Proc. IEEE Military Communications Conference (MILCOM)*, 2016.
- [25] E. Uysal, O. Kaya, A. Ephremides, J. Gross, M. Codreanu, P. Popovski, M. Assaad, G. Liva, A. Munari, T. Soleymani, B. Soret, and K. H. Johansson, “Semantic communications in networked systems: A data significance perspective,” *arXiv preprint arXiv:2103.05391*, 2021.
- [26] S. Guo, Y. Wang, N. Zhang, Z. Su, T. H. Luan, Z. Tian, and X. Shen, “A survey on semantic communication networks: Architecture, security, and privacy,” *IEEE Communications Surveys & Tutorials*, vol. 27, no. 5, pp. 2860–2894, 2025.
- [27] L. Liang, H. Ye, Y. Sheng, O. Wang, J. Wang, S. Jin, and G. Y. Li, “Large language models for wireless communications: From adaptation to autonomy,” *arXiv preprint arXiv:2507.21524*, 2025.
- [28] Z. Lu, Y. Zuo, H. Xu, W. Chen, X. He, J. Guo, and S. Jin, “Wirelessop-sagent: A benchmark and agent design for action assurance in wireless networks,” *arXiv preprint arXiv:2608.08277*, 2026.
- [29] J. Wang *et al.*, “Can LLM agents respond to disasters? benchmarking heterogeneous geospatial reasoning in emergency operations,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2026, oral; *arXiv:2605.11633*.
- [30] S. Javdani, Y. Chen, A. Karbasi, A. Krause, J. A. Bagnell, and S. Srinivasa, “Near optimal bayesian active learning for decision making,” in *Proceedings of the Seventeenth International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 33, 2014, pp. 430–438.
- [31] B. Huang, C. Lu, L. Leqi, J. M. Hernandez-Lobato, C. Glymour, B. Schölkopf, and K. Zhang, “Action-sufficient state representation learning for control with structural constraints,” in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 9260–9279.
- [32] A. Deshpande, L. Hellerstein, and D. Kletenik, “Approximation algorithms for stochastic boolean function evaluation and stochastic submodular set cover,” *arXiv preprint arXiv:1303.0726*, 2013.
- [33] L. Hellerstein, D. Kletenik, N. Liu, and R. T. Witter, “Adaptivity gaps for the stochastic boolean function evaluation problem,” *arXiv preprint arXiv:2208.03810*, 2022.
- [34] D. Gao, Z. Xu, A. Rawashdeh, P. Klasnja, and S. Murphy, “Active measuring in reinforcement learning with delayed negative effects,” in *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 300, 2026, pp. 3907–3915.